

Original Manuscript ID: Access-2026-38801

Original Article Title: “The Price of De-correlation: Quantifying the Agreement-Cost Trade-off in Heterogeneous Large Language Model Ensembles”

To: IEEE Access Editor

Re: Response to reviewers

Dear Editor,

Thank you for allowing a resubmission of our manuscript, with an opportunity to address the reviewers’ comments.

We are uploading (a) our point-by-point response to the comments (below) (response to reviewers, under “Author’s Response Files”), (b) an updated manuscript with yellow highlighting indicating changes (as “Highlighted PDF”), and (c) a clean updated manuscript without highlights (“Main Manuscript”).

The revision adds two substantial pieces of evidence. First, we now measure the correlation between agents’ errors directly rather than inferring it from label agreement, which is what the term “de-correlation” in our title asserts. Second, we ran a pre-registered replication of the entire protocol on a substantially different domain, 537 expert clinical-decision items, and report it whatever the outcome. Every number below is regenerated by the released code from the frozen data.

Best regards,

Samir Chincholikar
Robin Chawla

Reviewer #1, Concern #1

Reviewer comment:

Reduced agreement is not shown to improve accuracy or any decision-relevant outcome. Link the agreement-cost frontier to calibration, abstention, robustness, or tail-error reduction.

Author response:

- We agree that reduced agreement alone does not establish a downstream benefit. We therefore added a selective-prediction analysis using mean agent conviction. AURC is 0.553 for HOM, 0.568 for HET-LITE, and 0.554 for HET; the HOM-HET difference is -0.001 with a 95% equity-clustered interval of [-0.059, 0.060] and a two-way interval of [-0.113, 0.115].
- Restricting evaluation to the most confident half of decisions does not materially reduce selective risk, so we do not claim improved calibration or abstention quality.
- As a separate exploratory outcome, HOM produces unanimous incorrect directional decisions at 1.84 times the rate of HET, with equity-clustered and two-way 95% intervals of [1.42, 2.52] and [1.13, 3.06], respectively. Because this analysis is post hoc and unadjusted for multiplicity, we treat it as hypothesis-generating rather than confirmatory.

Author action:

- Section V-H, “Exploratory Analysis of Consensus Errors,” was revised to report the unanimous-incorrect ratio with equity-clustered and two-way confidence intervals and to state explicitly that the analysis is exploratory.
- Section V-D, “The Agreement-Cost Frontier,” was revised to state that the frontier quantifies the cost associated with greater behavioral diversity and does not by itself establish improved downstream decision quality.
- Section V-I, “Selective Prediction,” is now a standalone subsection rather than part of the clinical replication, since its equity-clustered and two-way intervals concern the financial experiment. It defines the decision unit, the HOLD and undefined-sign exclusions, the coverage denominator, the conviction ranking, selective risk, the trapezoidal AURC, the within-draw paired contrast, and the bootstrap, and it follows the risk-coverage framework of El-Yaniv and Wiener [19].
- Reference [19], El-Yaniv and Wiener on selective prediction, was added to the reference list as the source of the risk-coverage framework used in this analysis.

Reviewer #1, Concern #2

Reviewer comment:

The tier-matched control covers only two providers and no xAI equivalent. A fully tier-matched multi-provider design or model-level random effects is needed.

Author response:

- A fully three-provider tier-matched design remains unavailable because an equivalent efficient-tier xAI model was not available.
- We therefore added an approximate capability comparison in the clinical replication: one same-provider pair and one cross-provider pair have nearly identical mean member accuracy, 0.951 and 0.955, while their error correlations are 0.954 and 0.607, a difference of 0.347 (95% CI [0.207, 0.507]).

- This reduces the capability confound but does not eliminate confounding between provider and other model-specific characteristics; we therefore do not identify provider as an independent causal factor.

Author action:

- Section V-J, “Cross-Domain Replication: Clinical Knowledge,” is new in this revision and includes the approximate capability-matched same-provider versus cross-provider comparison.
- Section V-K, “Robustness: Capability-Matched Control,” retains the pre-registered financial capability control, which preserves approximately 94% of the primary agreement contrast while holding model tier approximately constant.
- Section VII, “Threats to Validity,” subsection “Heterogeneity vs. capability confound,” was revised to state explicitly that provider remains confounded with model identity, training data, architecture, alignment, and effective capability.

Reviewer #1, Concern #3

Reviewer comment:

Findings come from one financial task, one prompt, three labels, and fixed five-agent ensembles. Test other domains, ensemble sizes, aggregation rules, and label structures.

Author response:

- We added a pre-registered replication using 537 questions from the MMLU professional-medicine and clinical-knowledge subsets, changing the task from three BUY/HOLD/SELL labels to four-option clinical questions.
- The ordering reproduces: agreement decreases by $\Delta\kappa = 0.072$ (95% CI [0.040, 0.104]) and error correlation by $\Delta\phi = 0.393$ (95% CI [0.325, 0.459]), compared with 0.336 and 0.454 in finance.
- The agreement effect is therefore materially smaller across domains, and we do not claim magnitude invariance. Ensemble size, prompt format, aggregation rule, and model mixtures remain fixed and are stated as limitations.

Author action:

- Section V-J, “Cross-Domain Replication: Clinical Knowledge,” was added to report the pre-registered 537-item clinical replication.
- Section VII, “Threats to Validity,” subsection “Scope and generalizability,” was revised to distinguish the dimensions tested across domains from those that remain fixed.
- The clinical replication pre-registration and freeze receipt were added to the released reproducibility materials.
- Reference [20], Hendrycks et al., was added to the reference list as the source of the MMLU benchmark from which the clinical items are drawn.

Reviewer #1, Concern #4

Reviewer comment:

The 1.84x unanimous-incorrect finding and HET-LITE recommendation are post hoc, lack CIs/p-values, and

ignore multiplicity. Treat as hypothesis-generating only.

Author response:

- We now treat the consensus analysis explicitly as exploratory. The HOM-to-HET unanimous-incorrect ratio is 1.84, with an equity-clustered 95% interval of [1.42, 2.52] and a two-way interval of [1.13, 3.06].
- Because the statistic and configuration comparison were selected post hoc and no multiplicity adjustment is applied, these intervals are not interpreted as confirmatory evidence.
- HET-LITE is no longer presented as a recommended design; its apparent cost efficiency is treated only as a hypothesis for prospective confirmatory evaluation.

Author action:

- Section V-H, “Exploratory Analysis of Consensus Errors,” was revised to report equity-clustered and two-way uncertainty for the unanimous-incorrect ratio.
- Section V-H was also revised to state the post hoc status of the analysis, note the absence of multiplicity adjustment, and remove the HET-LITE design recommendation.
- Section VI, “Discussion,” was revised so that the HET-LITE result is described as hypothesis-generating rather than deployment guidance.

Reviewer #1, Concern #5

Reviewer comment:

Seed support is best-effort, and the diagnostic is strongest only for HOM. Heterogeneous cross-run checks are confounded by repeated same-model draws. Add verification or sensitivity analysis.

Author response:

- We added a pair-type sensitivity analysis. Same-model agreement is 0.800 in HOM, 0.794 in HET-LITE, and 0.862 in HET, while different-model agreement is 0.455 in HET-LITE and 0.490 in HET. The reduction in ensemble agreement is therefore associated with the introduction of different-model pairs rather than a systematic reduction among repeated instances of the same model.
- We also clarify that distinct per-agent seeds remove the specific shared-seed artifact observed in the pilot but do not establish statistical independence of model responses.

Author action:

- Section V-E, “Shared-Seed Dependence and Measurement Validation,” subsection “Sensitivity: agreement by pair type,” was added to report the same-model and different-model pairwise agreement results.
- Section V-E, subsection “What unique seeds do and do not establish,” was added to state explicitly that distinct per-agent seeds do not prove statistical independence.
- Figure 2 was revised to describe the change as assignment of a distinct reproducible seed to each agent and explicitly states that this does not establish statistical independence.
- Table 6 retains the within-run versus cross-run diagnostic and identifies the heterogeneous results as secondary consistency checks rather than direct independence tests.
- The Conclusion no longer refers to verifying sampling independence; it now describes checking for residual within-run dependence and avoiding shared-seed artifacts, matching Section III-C. The claim

that reproducing κ_{HOM} across datasets re-confirms determinism has been removed and replaced with a consistency check, since reproducing an aggregate statistic does not demonstrate determinism.

Reviewer #1, Concern #6

Reviewer comment:

Table 4 and the text disagree on costs. Clarify per-decision cost, token accounting, and pricing assumptions.

Author response:

- We clarified that per-call cost is computed from metered prompt and completion tokens using the model-registry rates, summed across the five agents within each ensemble decision, and then averaged across cells.
- The resulting costs are $\$6.48 \times 10^{-4}$ for HOM, $\$1.52 \times 10^{-3}$ for HET-LITE, and $\$2.84 \times 10^{-3}$ for HET, so HOM-to-HET increases inference cost by approximately 4.4x.
- Prices are fixed at those applicable during collection; no discount, batching, or caching adjustment is applied, and failed requests with no billable completion are excluded.

Author action:

- Section V-D, “The Agreement-Cost Frontier,” was revised to state the token accounting, aggregation, and pricing assumptions explicitly.
- Table 4 reports the corresponding per-ensemble-decision cost for each of the three configurations.

Reviewer #2, Concern #1

Reviewer comment:

Simplify index terms to 3 - 5 words only.

Author response:

- Done. The Index Terms have been reduced to five terms.

Author action:

- The manuscript Index Terms were revised to: ensemble learning, error correlation, large language models, model heterogeneity, and multi-agent systems. The five terms are listed alphabetically, as IEEE Access requires.

Reviewer #2, Concern #2

Reviewer comment:

No clear explanation about fair comparison with current state-of-the-art approach.

Author response:

- This study does not propose a new ensemble algorithm, so a leaderboard-style task-performance comparison with existing ensemble methods would address a different question. Prior work typically evaluates

mixture-of-agents, debate, or ensemble-selection methods through downstream performance, whereas we hold the task, prompt, information set, and aggregation rule fixed and study how ensemble composition relates to agreement, error dependence, and cost.

- The appropriate comparison is therefore methodological: our framework separates output agreement from error correlation, controls the shared-seed artifact, quantifies statistical uncertainty, and incorporates inference cost.

Author action:

- Section V-M, “Relation to Existing Ensemble Methods,” was added to explain the basis of comparison with existing ensemble approaches.
- Section II-A, “LLM Ensembles and Mixture-of-Agents,” provides the corresponding literature context for mixture-of-agents, diversity-based selection, and heterogeneous ensemble approaches.
- Section V-M, “Relation to Existing Ensemble Methods,” now states the distinction as one of estimand rather than quality: prior aggregation and selection work asks whether a method improves downstream performance, while this study holds the task, prompt, information set and aggregation rule fixed and asks how ensemble composition relates to agreement, error dependence and cost. It identifies the four methodological axes on which the comparison is appropriate, and cites Kuncheva and Whitaker [17] for the classifier-ensemble result that diversity and accuracy are related but not interchangeable, which is why reduced agreement is not treated here as evidence of improved accuracy.

Reviewer #3, Concern #1

Reviewer comment:

The present experiment confounds provider, model identity, capability, training data, architecture, and alignment. Add same-provider/different-model controls, more than one model per provider, balanced provider mixtures, and preferably a factorial analysis; otherwise rewrite all causal/provider-specific claims as configuration-specific observational findings.

Author response:

- We agree that provider is not experimentally separable from model identity, capability, training data, architecture, alignment procedure, and other model-specific characteristics in the primary design. We therefore interpret the comparisons as configuration-specific observational findings rather than isolated causal provider effects.
- The pre-registered financial capability control preserves approximately 94% of the primary agreement contrast under approximate tier matching, and the clinical replication provides an additional approximate capability-matched comparison with similar member accuracy but substantially different error correlation.
- These controls narrow the capability confound but do not identify provider as an independent causal factor.

Author action:

- Section V-F, “Provider-Pair Agreement,” was rewritten as a configuration-specific observational analysis and now states explicitly that provider is bundled with other model characteristics.
- Section V-J, “Cross-Domain Replication: Clinical Knowledge,” is new in this revision and includes the approximate capability-matched same-provider versus cross-provider comparison.

- Section V-K, “Robustness: Capability-Matched Control,” retains the pre-registered financial capability control.
- Section VII, “Threats to Validity,” subsection “Heterogeneity vs. capability confound,” was revised to state explicitly that provider cannot be isolated causally in the present design.
- Causal attribution to provider has been removed throughout rather than only in the provider-pair subsection. The Abstract, Introduction, the metrics definition, Section V-A, the Discussion and the Conclusion now describe an association across the evaluated configurations. The phrase “capability differences alone do not explain the observed contrast” has been replaced in all three places it appeared with the statement that the contrast remains substantial under approximate capability matching, although that control does not isolate provider from other model-specific characteristics.

Reviewer #3, Concern #2

Reviewer comment:

Recompute uncertainty while accounting for both repeated observations within equities and shared analysis dates/common market conditions. Justify the bootstrap unit formally, evaluate two-way equity/date clustering or an appropriate multiway resampling strategy, and regenerate all main and exploratory confidence intervals accordingly.

Author response:

- Equity remains the pre-registered primary resampling unit because observations are repeated within securities, while analysis date is a crossed dependence dimension because all securities evaluated on the same date share market conditions. We therefore independently resample equities and dates with replacement, preserving multiplicity on both margins.
- The primary HOM-HET contrast remains $\Delta\kappa = 0.3363$, with 95% intervals of [0.3035, 0.3689] under equity clustering, [0.2989, 0.3656] under date clustering, and [0.2786, 0.3877] under two-way resampling. All pre-registered secondary and exploratory inferential contrasts were regenerated under the appropriate scheme, and no substantive conclusion changes.
- The temperature analysis remains equity-clustered because its two analysis dates are insufficient for two-way inference; the clinical replication instead resamples MMLU items because the financial equity-date structure does not apply.
- During this revision we also identified and corrected an implementation that had collapsed repeated sampled equities to a set, thereby losing bootstrap multiplicity and understating variance. The corrected code preserves multiplicity on both margins and now asserts this property automatically.

Author action:

- Section III-D, “Uncertainty Quantification,” justifies equity as the primary cluster, identifies date as the crossed dimension, and defines the date-clustered and two-way procedures.
- Table 5 reports the primary contrast under equity, date, and two-way resampling.
- Section V-A, “Heterogeneity and Ensemble Agreement (Primary),” now reports the two-way intervals for both pre-registered secondary contrasts alongside Table 2.
- Section V-K, “Robustness: Capability-Matched Control,” now reports the two-way interval for the capability-matched contrast.
- Section V-C, “Agreement, Error Correlation, and Decorrelation,” reports two-way intervals for both

error-correlation specifications.

- Section V-H, “Exploratory Analysis of Consensus Errors,” reports two-way intervals for the unanimous-incorrect ratio and the pile-on contrast.
- Section V-I, “Selective Prediction,” reports the two-way AURC interval.
- Section V-L, “Robustness: Temperature Sensitivity,” states that the subgrid spans two analysis dates, which is too few clusters for date-clustered or two-way inference, and that the analysis therefore contributes no independent multiway evidence.
- Section III-D now gives the methodological precedent for the crossed procedure: the pigeonhole bootstrap [14], which resamples the two margins separately, and the multiway-inference literature for non-nested clustering dimensions [15]. We state explicitly that our procedure is not an implementation of a multiway variance estimator.
- Table 5 previously labelled the two-way row “1200 clusters”, which can be read as 1,200 independent clusters. It now reads “100 x 12” under a “Resampled units” column. The point estimate and interval are unchanged.
- References [14], Owen’s pigeonhole bootstrap, and [15], Cameron, Gelbach and Miller on multiway clustering, were added to the reference list as the methodological precedent for the crossed resampling procedure.

Reviewer #3, Concern #3

Reviewer comment:

Fleiss’ kappa measures agreement, not correlation. Either remove “de-correlation”/“less-correlated judgments” throughout the title, abstract, figures, discussion, and conclusion, or add actual dependence/error-correlation analyses that justify those terms. Explicitly distinguish output agreement, error correlation, sampling independence, model diversity, and provider diversity.

Author response:

- We took the reviewer’s second option and now measure error dependence directly using ϕ correlations between paired binary error indicators. Error correlation decreases from 0.984 for HOM to 0.742 for HET-LITE and 0.530 for HET, giving a HOM-HET contrast of 0.454 (95% CI [0.376, 0.534]; two-way [0.328, 0.586]).
- Counting HOLD as an error to hold the evaluated subset fixed gives the same ordering, with a HOM-HET contrast of 0.309 (95% CI [0.262, 0.354]; two-way [0.227, 0.387]). Mean per-agent error rates remain similar, so the reduction reflects lower error dependence rather than differences in average correctness.
- The manuscript now explicitly distinguishes output agreement, error correlation, sampling independence, model diversity, and provider diversity; “de-correlation” refers specifically to reduced error correlation measured by ϕ , while κ measures categorical agreement.

Author action:

- Section V-C, “Agreement, Error Correlation, and Decorrelation,” was added to define the five concepts and report the ϕ analysis under both treatments of HOLD.
- Section V-A was retitled “Heterogeneity and Ensemble Agreement (Primary)” so that the κ result is named as an agreement result and the heading itself carries no causal verb.
- The Abstract was revised to report the ϕ contrast alongside the κ contrast and to distinguish reduced

error dependence from reduced agreement.

- Section VI, “Discussion,” and Section VIII, “Conclusion,” were revised so that decorrelation claims refer to error dependence rather than categorical agreement. The Conclusion now states that model diversity can reduce both output agreement and error correlation.
- Section V-C now states the phi computation exactly as implemented: within each (configuration, equity, date, run) cell we form, for every unordered pair of agent positions, the pair of binary error indicators against the realized forward-return sign; those pairs are pooled over all cells and all agent-position pairs within a configuration, and phi is the Pearson correlation of the pooled set, so one phi is reported per configuration rather than one per agent pair. The previous wording implied a correlation per pair, which a single paired observation cannot support.
- Reference [17], Kuncheva and Whitaker on the relationship between ensemble diversity and accuracy, was added to the reference list to support the separation of reduced error dependence from improved accuracy.

Reviewer #3, Concern #4

Reviewer comment:

HOLD-induced abstention creates different evaluation subsets across configurations, so the reported hit rates are not directly comparable. Perform a proper selective-prediction/coverage analysis, use paired and clustered inference, provide uncertainty for unanimous-error and pile-on statistics, and stop using overlapping marginal Wilson intervals as the main inferential evidence.

Author response:

- We added selective-prediction analysis with paired clustered inference. The HOM-HET AUC difference is -0.001, with an equity-clustered 95% interval of [-0.059, 0.060] and a two-way interval of [-0.113, 0.115], both containing zero.
- We also report uncertainty for the exploratory consensus outcomes: the unanimous-incorrect ratio is 1.84 with intervals [1.42, 2.52] and [1.13, 3.06], while the pile-on difference is 0.189 with intervals [0.160, 0.216] and [0.140, 0.232] under equity and two-way resampling, respectively.
- As a common-subset robustness check, counting HOLD as an error yields a HOM-HET ϕ contrast of 0.309 with equity and two-way intervals of [0.262, 0.354] and [0.227, 0.387]. Wilson intervals are now retained only as descriptive summaries.

Author action:

- Section V-I, “Selective Prediction,” was added to report risk-coverage analysis and paired clustered AUC inference.
- Section V-H, “Exploratory Analysis of Consensus Errors,” was revised to report equity-clustered and two-way uncertainty for unanimous incorrect decisions and pile-on.
- Section V-C, “Agreement, Error Correlation, and Decorrelation,” is new in this revision and reports the common-subset error-correlation analysis with HOLD counted as an error.
- Section V-G, “Secondary Directional-Accuracy Analysis,” retains the directional hit-rate analysis as secondary and does not use overlapping marginal Wilson intervals as the principal inferential evidence.

Reviewer #3, Concern #5

Reviewer comment:

Clarify that unique seeds do not prove statistical independence; provide complete API parameters and versions, exact prompts/system messages, seed-support evidence, retry records, total attempted versus successful API calls, exact preregistration timestamp evidence, and archived immutable code/data. Correct Figure 3 so that the shared-seed pilot, independent-seed pilot, and full-study quantities compare the same estimand.

Author response:

- We now state explicitly that distinct per-agent seeds remove the demonstrated shared-seed artifact but do not establish statistical independence.
- The confirmatory grid contains 54,000 valid outputs and required 94,566 API requests; of the 40,566 unsuccessful requests, 40,544 were rate-limit or quota responses and 22 were transport or schema failures. Every attempt is retained in the released log together with request settings, prompts, model identifiers, seeds, prices, archived responses, and frozen analysis materials.
- Figure 3 now compares only like-for-like pilot quantities on the same grid and model set: the HOM-HET agreement contrast is 0.485 under a shared seed and 0.113 under distinct per-agent seeds. The full-study within-minus-cross-run diagnostic is reported separately in Table 6.
- The released package also contains timestamped cryptographic freeze receipts for both the confirmatory study and the cross-domain replication, with the latter recorded before any API call for that arm.

Author action:

- Section III-C, “Independent-Draw Estimand,” was revised to document the per-agent seed construction and the limits of seed-based reproducibility.
- Section IV-C, “Models,” records the model identifiers, provider assignments, inference settings, timestamps, seeds, prompt hashes, token usage, costs, and archived responses.
- Section IV-D, “Pre-Registration, Freezing, and Reproducibility,” documents the frozen design, pre-registration hash, immutable dataset, and deterministic analysis pipeline.
- Section IV-E, “Collection Provenance and Retry Accounting,” was added to report the 94,566 attempted requests, 54,000 valid outputs, failure breakdown, and retained attempt history.
- Section V-E, “Shared-Seed Dependence and Measurement Validation,” was revised to distinguish removal of the shared-seed artifact from proof of statistical independence.
- Figure 2 was revised to show the distinct per-agent seed construction and explicitly state that this does not establish statistical independence.
- Figure 3 was reduced to the two directly comparable pilot HOM-HET agreement contrasts.
- Table 6 reports the full-study within-run versus cross-run diagnostic separately.
- Section IV-D now identifies the freeze receipts in the released package. The confirmatory freeze is timestamped 2026-07-03T20:41:51.396959+00:00 and records the SHA-256 of the configuration, agent and analysis code, prompt template, pre-registration, grid, and model manifest together with the commit that carried them. The cross-domain replication has its own receipt timestamped 2026-09-29T14:47:54.287335+00:00, recorded before any API call for that arm.

Reviewer #3, Concern #6

Reviewer comment:

Replicate the main experiment on at least one substantially different domain/task and preferably multiple prompt formats, ensemble sizes, and model mixtures. The present single BUY/HOLD/SELL financial task-whose accuracy is itself near chance and whose label marginals vary strongly by configuration-is insufficient to establish a general principle about heterogeneous LLM ensembles.

Author response:

- We added a pre-registered replication using 537 MMLU professional-medicine and clinical-knowledge questions with four response options rather than the three financial labels.
- The main ordering reproduces: agreement decreases by $\Delta\kappa = 0.072$ (95% CI [0.040, 0.104]) and error correlation by $\Delta\phi = 0.393$ (95% CI [0.325, 0.459]). The agreement contrast is substantially smaller than in finance, so we do not claim magnitude invariance across tasks.
- Prompt format, five-agent ensemble size, majority-vote aggregation, and the tested model mixtures remain fixed and are explicitly stated as limitations.

Author action:

- Section V-J, “Cross-Domain Replication: Clinical Knowledge,” was added to report the pre-registered 537-item clinical replication.
- Section VII, “Threats to Validity,” subsection “Scope and generalizability,” was revised to state that the ordering has been evaluated across two substantially different tasks while prompt format, ensemble size, aggregation rule, and model mixtures remain fixed.
- The clinical replication pre-registration and freeze receipt were added to the released reproducibility materials.
- Reference [20], Hendrycks et al., was added to the reference list as the source of the MMLU benchmark used for this replication.